Exam 3 Practice Problems — Answers

Equivalent mathematical forms are valid. In planar flux questions, the normal is outward.

1. (a) $\nabla \times \mathbf{F} = (8y + 3z^2, 0, 3 - x^2)$, $\nabla \cdot \mathbf{F} = 2xy$. Not conservative.

   (b) $\nabla \times \mathbf{F} = \left(\dfrac{x^3y^2}{z^2}, 4z - 7, \dfrac{3x^2y^2}{z} \right)$, $\nabla \cdot \mathbf{F} = 2 + \dfrac{2x^3y}{z}$. Not conservative on either half-space $z > 0$ or $z < 0$.

   (c) $\nabla \times \mathbf{F} = (0, 3x^2, 0)$, $\nabla \cdot \mathbf{F} = 14 - 2y$. Not conservative.

   (d) $\nabla \times \mathbf{F} = (0, 0, 0)$, $\nabla \cdot \mathbf{F} = -xz \cos y$. Conservative, with potential $f = xz \cos y + C$.

2. Conservative: $\nabla \times \mathbf{F} = \mathbf{0}$ and $\boxed{f = x^2y^2 + xz + C}$. Indeed, $P_y = Q_x = 4xy$, $P_z = R_x = 1$, and $Q_z = R_y = 0$.

3. Conservative, with $\boxed{f = xyz + e^x + e^y + e^z + C}$.

4. $\mathbf{F} = \nabla(xyz^2)$, so the integral is $f(1,1,1) - f(0,0,0) = \boxed{1}$.

5. $\boxed{3e - 2}$, using the potential from Problem 3.

6. $Q_x - P_y = -1 - 2y$, so
$$\int_0^1 \int_0^x (-1 - 2y) \, dy \, dx = \boxed{-\frac{5}{6}}.$$

7. $P_x + Q_y = 0$, so the outward flux is $\boxed{0}$.

8. $\boxed{-8\pi}$.

9. $\boxed{8\pi}$.

10. $\boxed{1}$.

11. $\boxed{8}$.

12. $Q_x - P_y = y$, so
$$\int_0^1 \int_{x^2}^x y \, dy \, dx = \frac{1}{2} \int_0^1 (x^2 - x^4) \, dx = \boxed{\frac{1}{15}}.$$

13. $P_x + Q_y = 3x$, so
$$\int_0^1 \int_{x^2}^x 3x \, dy \, dx = \boxed{\frac{1}{4}}.$$

14.

$$\mathbf{r}_u \times \mathbf{r}_v = (2u^2 + 2v^2, v - u, -2(u + v)),$$

$$\boxed{A = \int_0^2 \int_0^1 \sqrt{(2u^2 + 2v^2)^2 + (v - u)^2 + 4(u + v)^2}\ \mathrm{d}u\ \mathrm{d}v}.$$

15.

$$\mathbf{r}_u \times \mathbf{r}_v = (v^2 - 2uv, u^2 - 2uv, 3u^2v^2),$$

$$\boxed{m = \int_0^1 \int_0^1 e^{u+v} \sqrt{(v^2 - 2uv)^2 + (u^2 - 2uv)^2 + 9u^4v^4}\ \mathrm{d}u\ \mathrm{d}v}.$$

16.

$$\boxed{A = 2\pi \int_0^1 e^{5x} \sqrt{1 + 25e^{10x}}\ \mathrm{d}x}.$$

17. Interpreting the scalar function on the spatial surface as $f(x, y, z) = y^5$,

$$\boxed{\iint_S f\ \mathrm{d}\sigma = 2\pi \int_1^4 x^{-3/2} \sqrt{1 + \frac{1}{4x^3}}\ \mathrm{d}x}.$$

The radius is x, the density is $(x^{-1/2})^5$, and the generating curve has speed $\sqrt{1 + 1/(4x^3)}$.

18. $\boxed{\pi/3}$.

19. Use $\mathbf{r}(r, \theta) = (r\cos\theta, r\sin\theta, r)$, with $2 \le r \le 4$, $0 \le \theta \le 2\pi$. Then

$$\iint_S z^2\ \mathrm{d}\sigma = \int_0^{2\pi} \int_2^4 r^2 \sqrt{2}\, r\ \mathrm{d}r\ \mathrm{d}\theta = \boxed{120\sqrt{2}\,\pi}.$$

20.

$$A = \int_0^{2\pi} \int_0^1 r\sqrt{1 + 4r^2}\ \mathrm{d}r\ \mathrm{d}\theta = \boxed{\frac{\pi(5\sqrt{5} - 1)}{6}}.$$

21. The spherical band has $\pi/4 \le \phi \le 3\pi/4$, so

$$A = \int_0^{2\pi} \int_{\pi/4}^{3\pi/4} \sin\phi\ \mathrm{d}\phi\ \mathrm{d}\theta = \boxed{2\sqrt{2}\,\pi}.$$

22. $\boxed{3\sqrt{14}}$.

23. With $\mathbf{r}(\phi,\theta) = (\sin\phi\cos\theta, \sin\phi\sin\theta, \cos\phi)$, the outward vector area element is $\sin\phi\,\mathbf{r}\,\mathrm{d}\phi\,\mathrm{d}\theta$. The complete setup is

$$\boxed{\int_0^{2\pi}\int_0^{\pi}\left(\sin^3\phi\cos^2\theta\sin\theta + 2\sin\phi\sin\theta - \cos^2\phi\right)\sin\phi\;\mathrm{d}\phi\;\mathrm{d}\theta}.$$

24. The downward vector area element is $(r\cos\theta, r\sin\theta, -r)\;\mathrm{d}r\;\mathrm{d}\theta$, giving

$$\int_0^{2\pi}\int_2^4 (r^2\sin\theta\cos\theta - r)\;\mathrm{d}r\;\mathrm{d}\theta = \boxed{-12\pi}.$$

25. The upward graph normal factor is $(1/3, 1, 1)$, and $(3, 1, -2)\cdot(1/3, 1, 1) = 0$. Thus the flux is $\boxed{0}$.

26. The plane is $z = 6 - 3x/5 - 2y/5$, and the upward flux integrand is $(3y + 7x)/5$ over the triangle with vertices $(0,0), (10,0), (0,15)$:

$$\int_0^{10}\int_0^{15-3x/2}\frac{3y+7x}{5}\;\mathrm{d}y\;\mathrm{d}x = \boxed{575}.$$

27. With $z_x = 2xy^2 + 1$ and $z_y = 2x^2y + 1$, the setup is

$$\boxed{\int_0^1\int_0^1 (-xy^2 - 4x^2y - 3)\;\mathrm{d}x\;\mathrm{d}y}.$$

Its value, for a check, is $-23/6$.

28. With $z_x = 4x^3$ and $z_y = 2y$, the setup is

$$\boxed{\int_{-1}^1\int_{-1}^1 \left(-4x^5 - 2y(x^4+y^2) + (x^4+y^2)^2\right)\;\mathrm{d}x\;\mathrm{d}y}.$$

Its value, for a check, is $16/9$.

29. $\boxed{-\pi}$. The intended surface includes its equatorial rim ($z \geq 0$); the rim is the boundary used in Stokes' theorem.

30. $\boxed{0}$.

31. $\boxed{0}$.

32. $\boxed{-5\pi}$.

33. $\boxed{0}$.

34. $\boxed{0}$. The hemisphere together with its base disk is closed, so its boundary is empty.

35. $\boxed{0}$. The entire cube surface is closed.

36. Valid parametrizations and bounds are:
    (a) $\mathbf{r}(u,v) = (u, v, 10)$, $-1 \le u \le 1$, $-2 \le v \le 2$. The stated bounds describe a rectangle, rather than a square.
    (b) $\mathbf{r}(u,v) = (u, v, u^2 + v^2)$, $0 \le v \le 1$, $0 \le u \le v$.
    (c) $\mathbf{r}(r,\theta) = (r\cos\theta, r\sin\theta, r^2)$, $0 \le r \le 3$, $0 \le \theta \le 2\pi$.
    (d) $\mathbf{r}(r,\theta) = (r\cos\theta, r\sin\theta, r)$, $2 \le r \le 4$, $0 \le \theta \le 2\pi$.
    (e) $\mathbf{r}(r,\theta) = (r\cos\theta, r\sin\theta, 4 - r\cos\theta + r\sin\theta)$, $0 \le r \le 3$, $0 \le \theta \le 2\pi$.
    (f) $\mathbf{r}(\theta,z) = (\cos\theta, \sin\theta, z)$, $0 \le \theta \le \pi/2$, $0 \le z \le 3$.
    (g) $\mathbf{r}(\theta,z) = (2\cos\theta, 2\sin\theta, z)$, $0 \le \theta \le 2\pi$, $2\sin\theta - 2 \le z \le 2\cos\theta + 4$.
    (h) $\mathbf{r}(\phi,\theta) = (2\sin\phi\cos\theta, 2\sin\phi\sin\theta, 2\cos\phi)$, $0 \le \phi \le \pi/4$, $0 \le \theta \le 2\pi$.

37. $\boxed{\text{(a), (d), (e)}}$ under the course convention that ∇ denotes the gradient of a *scalar* field; a Jacobian of a vector field is not being requested in option (c).

38. $\boxed{\text{(a), (b), (d)}}$.

39. $\boxed{\text{(a), (b), (d)}}$.

40. In order: $\boxed{\text{F, T, T, F, F, T}}$.

41. Choices are (a) direct parametrization, (b) FTLI, (c) Green, (d) Stokes, (e) none. The coordinator's intended *direct-method* selections are listed below. The original question does not explicitly exclude planar reductions; those also make Green valid in (3) and (5).

Case	Direct selections	Allowing planar Green	Integral value
(1)	(a), (c), (d)	same	2π
(2)	(a), (b), (c), (d)	same	0
(3)	(a), (d)	(a), (c), (d)	0
(4)	(a), (b)	same	2
(5)	(a), (d)	(a), (c), (d)	$-3/2$
(6)	(a), (b), (c), (d)	same	0

In (2), (4), and (6), potentials are x^2y, x^2y+xz, and x^3y+y^2, respectively. In (4) the curve is open. In (1) the planar scalar curl is 2. The three-dimensional curls in (3) and (5) are $(1,1,1)$ and $(-1,-1,-1)$, respectively, so neither field is conservative.

For (3), substitute $z = y + 2$, $\mathrm{d}z = \mathrm{d}y$ into the line integral:

$$\mathbf{F}\cdot\mathrm{d}\mathbf{r} = (y+2)\,\mathrm{d}x + (x+y)\,\mathrm{d}y, \qquad Q_x - P_y = 0.$$

For (5), substitute $z = 1 - x - y$, $\mathrm{d}z = -\mathrm{d}x - \mathrm{d}y$:

$$\mathbf{F}\cdot\mathrm{d}\mathbf{r} = (y-x)\,\mathrm{d}x + (1-2x-y)\,\mathrm{d}y, \qquad Q_x - P_y = -3.$$

Thus both curves are planar (in tilted planes), and Green can be applied after these explicit reductions. The base triangle in (5) has area 1/2.